



ANALOGUE CAPACITANCE METER

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The principle of operation of this meter is based on pulse-width modulation. Larger the capacitance, higher is the pulse-width and vice-versa. The meter given here has one oscillator and two one-shot generators (OSG) built around RC components. An OSG is triggered by a free-running oscillator built around IC NE555 (IC1). The pulse-width of the OSG is determined by the capacitor under measurement (C_x). By passing the pulses from the OSG through a linear ammeter it is possible to measure its duty cycle. In other words, the deflection of the ammeter can be made equal to the duty cycle, which here becomes proportional to the value of the capacitor C_x due to linear property of the OSG time period.

Fig. 1 shows the circuit diagram of analogue capacitance meter. The OSG comprises timing resistors (R4

through R6) and capacitor C_x , whose voltage is sampled by CMOS gate N1. C_x is discharged through transistor T3 connected in parallel. C_x is connected to pin 2 of gate N1. The voltage at pin 2 of gate N1 is used as the reference voltage to which C_x has to charge. Once this voltage is reached, the sampling gate output makes the transition from high to low state, indicating the completion of the charging time. Charging time (on-time) of capacitor C_x is given by:

$$T_{on} = RC \times \ln \left(\frac{V_o}{V_o - V_{th}} \right)$$

Here, V_o is supply voltage and V_{th} is the threshold input voltage of gate N1. $R=R4$ through R6 and $C=C_x$. Due to this linearity of the RC circuit, a capacitor with twice the value would take twice the time. Thus the OSG pulse-width, and hence the duty cycle of the output waveform, becomes pro-

portional to the value of the capacitor.

Due to stray capacitance of connecting wires the OSG does not produce a zero pulse without capacitor C_x . To eliminate this, a similar auxiliary OSG, comprising presets VR1 through VR3 and a capacitor C3, is used. The presets VR1 through VR3 are adjusted such that for each range the meter gives zero indication when capacitor C_x is not connected. With these adjustments the meter is ready for use.

The oscillator built around IC1 should be configured such that its on-time period (T_{on}) slightly exceeds the OSG on-time period (T_{on}) with the full range capacitor value and off-time period (T_{off}) occurs just before complete discharge of capacitor. This becomes very important for capacitors having large capacitances.

The calibration of the ammeter is easy if suitable reference capacitor is available. It is recommended to per-

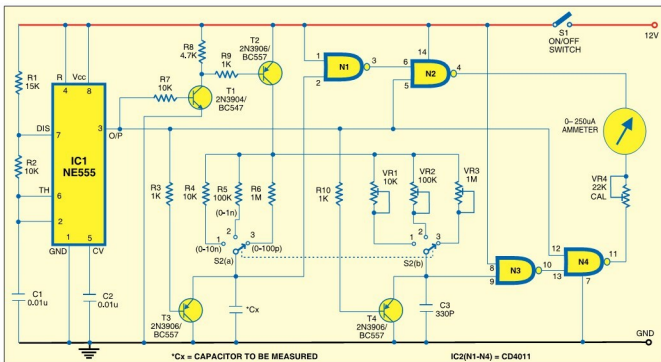


Fig. 1: Circuit diagram of analogue capacitance meter